Activity A1 - Build the Environmental Baseline and Aspect Inventory

Course: Fundamentals of ISO 14001 Environmental Management System (EMS) (TGS-2023037592) Ability: A1 Objective: Identify workplace environmental hazards and aspects from process and site evidence. Mode: Team analysis with individual evidence submission Suggested time: 60 minutes

Scenario

MEP is preparing its first ISO 14001:2026-aligned EMS. Managers have energy bills and waste invoices, but no shared view of where activities interact with air, water, land, resources or neighbours.

Meridian EcoPack Pte Ltd (MEP) is a fictional Singapore SME with 120 employees producing printed corrugated and fibre-based packaging for regional food, electronics and healthcare customers. Two key customers require evidence of an ISO 14001:2026-aligned EMS and stronger value-chain environmental controls before contract renewal.

Task and expected output

Trace MEP's materials, energy, water, emissions, waste and emergency interfaces and distinguish hazard, aspect and impact.

Produce: A process-boundary map, environmental baseline and aspect-impact inventory with evidence references.

Before you begin

- Open a working copy of the Excel workbook; do not overwrite the supplied evidence.
- Allocate a facilitator, evidence recorder, checker and presenter.
- Read every evidence card before making a compliance or significance conclusion.
- If the activity concerns an incident, this is a tabletop exercise: do not handle chemicals or simulate unsafe physical actions.

Decision questions

1. Which activities, products and services create direct or influenced environmental interactions?
2. For each interaction, what is the aspect and what is the resulting environmental impact?
3. What normal, abnormal and emergency conditions must be included, and what evidence is still missing?

Detailed procedure

Step 1 - Read the company profile, process flow, site plan and evidence cards before scoring anything

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Record the evidence used, the decision made, the responsible owner and any uncertainty in the workbook.

Step 2 - Define the review boundary from receipt of paper and inks through printing, conversion, storage, dispatch and end-of-life influence

Define the review boundary from receipt of paper and inks through printing, conversion, storage, dispatch and end-of-life influence.

Record the evidence used, the decision made, the responsible owner and any uncertainty in the workbook.

Step 3 - Walk each process stage and record inputs, outputs, releases, resource use, waste streams and potential emergencies

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Step 4 - Write each aspect as a cause or interaction and each impact as a change to the environment

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Record the evidence used, the decision made, the responsible owner and any uncertainty in the workbook.

Step 5 - Tag the operating condition, control/influence level, evidence source, owner and missing data for every entry

Tag the operating condition, control/influence level, evidence source, owner and missing data for every entry.

Record the evidence used, the decision made, the responsible owner and any uncertainty in the workbook.

Step 6 - Peer-review for omissions at maintenance, contractors, utilities, transport and product end-of-life stages

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Record the evidence used, the decision made, the responsible owner and any uncertainty in the workbook.

Evidence to submit

- Completed Activity-A1-Workbook.xlsx.
- One evidence-backed team decision and one documented peer challenge.
- Any assumption or missing source clearly labelled for follow-up.

Acceptance checklist

[ ] The inventory covers normal, abnormal and potential emergency conditions; every aspect-impact pair is logically linked and traceable to supplied evidence or an explicit evidence gap.

[ ] Claims are traceable to supplied or cited evidence.

[ ] Assumptions, uncertainty and missing evidence are labelled.

[ ] Owners, due dates and next decisions are included where applicable.

[ ] Legal/compliance interpretations needing competent advice are escalated rather than guessed.

[ ] A peer challenge and response are recorded.

Troubleshooting

- The team cannot agree on a score: return to the criterion anchors and record each evidence-based assumption before averaging or escalating.
- Evidence is missing: do not invent it; record an evidence gap, risk implication, owner and retrieval action.
- A legal requirement is unclear: record the current authoritative source and escalate to a competent adviser or authority channel.

- A conclusion feels obvious: test an alternative explanation and state what evidence would falsify the current view.

Facilitator debrief

Ask each team to defend one priority, control or conclusion using objective evidence. Probe what would change the decision, then connect the work to PDCA and the relevant EMS process rather than clause memorisation.

Files in this activity pack

- README.md and README.pdf - detailed learner procedure.
- Activity-A1-Scenario-and-Checklist.pdf - printable scenario and completion checklist.
- Activity-A1-Workbook.xlsx - editable analysis and submission workbook.

Source and usage note

This fictional training activity paraphrases ISO 14001 concepts and is not legal advice or a reproduction of the standard. Use the organisation's licensed ISO 14001:2026 copy and current authoritative Singapore sources for formal interpretation.